

IMSE 774 · Neural Networks

In-Class Review · Chapters 1–4 · ANSWER KEY (instructor copy, do not distribute)

Fall 2026 · North Dakota State University

Coverage. Ch. 1: 5 · Ch. 2: 6 · Ch. 3: 10 · Ch. 4: 9 · 30 questions · 75 minutes.

Letters only. 1-b 2-b 3-c 4-b 5-a 6-a 7-a 8-a 9-c 10-a 11-d 12-c 13-b 14-b 15-d 16-b 17-d 18-a 19-d 20-a 21-d 22-b 23-c 24-c 25-d 26-c 27-c 28-d 29-a 30-b

Answer distribution. a: 8 · b: 8 · c: 7 · d: 7

Key with rationale

#	Ch.	Answer	Why	Reference
1	1	b	AI contains ML (logic and search are AI but not ML); ML contains deep learning, which is the procedure of fitting deep networks.	Prince §1, p. 1; fig. 1.1
2	1	b	No output labels; the goal is to describe structure in the inputs. The others are regression, binary classification, and reinforcement learning.	Prince §1.2, pp. 7–11
3	1	c	A probability over $N > 2$ discrete categories.	Prince §1.1.1, pp. 2–4; fig. 1.2
4	1	b	The data manifold is tiny compared with the input space; the network maps simple latents to complex data.	Prince §1.2.2, pp. 9–10; fig. 1.10
5	1	a	Rewards may arrive long after the action that earned them, and the agent must choose between exploiting known-good actions and exploring.	Prince §1.3, pp. 11–13
6	2	a	Training searches for the parameters that minimize this scalar; the test set is evaluated afterwards.	Prince §2.2.2, p. 19
7	2	a	Overfitting is a failure to generalize; the first option describes underfitting.	Prince §2.2.4, p. 21; Week 1 note
8	2	a	Setting it to zero gives the normal equations $\mathbf{X}^T \mathbf{X} \phi = \mathbf{X}^T \mathbf{y}$. Only this option has the right shape.	Week 2 note §8.3; UDL problem 2.1
9	2	c	Rows index outputs, columns index inputs: (outputs, inputs).	Prince App. B.3; Week 2 note §6
10	2	a	Apply $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ twice; only this ordering has compatible shapes.	Prince App. B.3.2

#	Ch.	Answer	Why	Reference
11	2	d	The condition number of the Hessian sets how many steps are needed; with a centered, unit-variance column it is 1.	Week 2 note §8.5; A1 task C8
12	3	c	The joint is where the unit's pre-activation $\theta_{d0} + \theta_{d1}x$ crosses zero.	Prince §3.1.1, pp. 26–27
13	3	b	Inactive units are clipped to zero and contribute nothing; ϕ_0 shifts the function vertically but never changes slope.	Prince §3.1.1, p. 27; UDL problem 3.9
14	3	b	The theorem is an existence result; it is silent on capacity, training, and generalization.	Prince §3.2, pp. 29–30; Week 3 note
15	3	d	A linear function of a linear function is linear: the hidden layer collapses and the 32 units buy nothing.	Prince §3.4, p. 35; UDL problem 3.1
16	3	b	Joints are set by the hidden layer alone; the output weights only choose slopes and offsets.	Prince §3.3.1, pp. 30–31; fig. 3.6
17	3	d	Three lines in general position cut the plane into $1 + 3 + \binom{3}{2} = 7$ cells; not all $2^3 = 8$ on/off patterns are geometrically realizable.	Prince §3.3.2, pp. 31–33; UDL problem 3.18
18	3	a	$D(D_i + 1) + D_o(D + 1) = 3 \cdot 4 + 2 \cdot 4 = 20$: fifteen weights and five biases.	Prince §3.4, p. 35; fig. 3.11; UDL problem 3.11
19	3	d	Activation is the value after the ReLU; pre-activation is before.	Prince §3.5, pp. 35–36; fig. 3.12
20	3	a	Saturation kills the gradient signal; the ReLU has its own problem (dying units on the negative side), so the last option is false.	Prince Ch. 3 notes, pp. 37–38; fig. 3.13
21	3	d	Non-negative homogeneity of the ReLU; every minimizer comes with a continuum of equivalent ones (and permuted copies).	UDL problems 3.5–3.7
22	4	b	The second network's three regions are copied once per region of the first: $3 \times 3 = 9$, against at most 7 for a shallow network with six units.	Prince §4.1, pp. 41–43; fig. 4.1
23	4	c	$\psi_{ij} = \theta'_{i1}\phi_j$: the first layer's three units are squeezed through the single number y . A general two-layer network has no such restriction.	Prince §4.2, pp. 43–45
24	4	c	Hyperparameters fix the family of functions; parameters select a member of it.	Prince §4.3.1, p. 46
25	4	d	$\mathbf{\Omega}_0$ acts on the input and $\mathbf{\Omega}_K$ produces the output, so $K + 1$ in all; rows for the layer produced, columns for the layer consumed.	Prince §4.4.1, pp. 48–49; fig. 4.6
26	4	c	$6(4 + 1) + 4(6 + 1) + 3(4 + 1) = 30 + 28 + 15 = 73$. (60 is the weights alone.)	Prince §4.4.1; UDL problem 4.4
27	4	c	Parameters $3D + 1 + (K - 1)D(D + 1) = 471$; regions up to $(D + 1)^K$. Regions grow exponentially in depth, parameters only linearly.	Prince §4.5.2, p. 50; fig. 4.7
28	4	d	Telgarsky (2016); Eldan & Shamir (2016). The reverse, width efficiency, is only polynomial.	Prince §4.5.3, p. 50; notes p. 53

#	Ch.	Answer	Why	Reference
29	4	a	The first is a theorem, the second a counting argument, the third a theorem (depth efficiency). Prince: ‘neither of these phenomena are well understood’ (p. 51).	Prince §4.5.5, p. 51
30	4	b	Representability and trainability are different questions; Chapters 6–7 and 11 address the second.	Week 4 note, ‘Deep Networks in PyTorch’; Prince §4.5.5

Using the review in class

The questions map onto the learning objectives of the Week 1–4 notes, so a per-question tally after the review shows which objectives to revisit before the midterm on October 22. Items designed to expose specific gaps: the shape of the least-squares gradient (Q8), the Zaslavsky count and why 2^D is only a bound (Q17), the rank-one connection between composition and a two-layer network (Q23), and the distinction between representability and trainability (Q30).

Suggested pacing for the 75-minute slot: 55 minutes for the paper, then 20 minutes going over the items with the lowest scores. Students who finish early may be asked to write one sentence per chapter naming the idea they found least clear; those sentences are worth collecting.